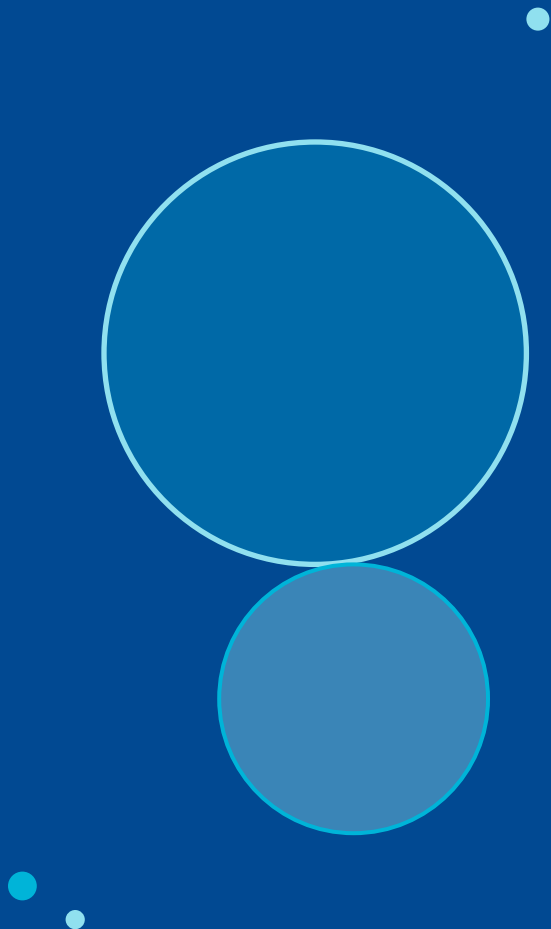


SkillClaw

Collective Skill Evolution for
Multi-User LLM Agent Ecosystems

Let Skills Evolve Collectively with Agentic Evolver

arxiv.org/abs/2604.08377



Agent Skills Remain Static After Deployment

Users Rediscover Solutions Independently

Static Skills (Current)

- Manually installed from hub
- Never updated from runtime experience
- Solutions lost after each session
- Same failures recur across users
- No system-level knowledge accumulation



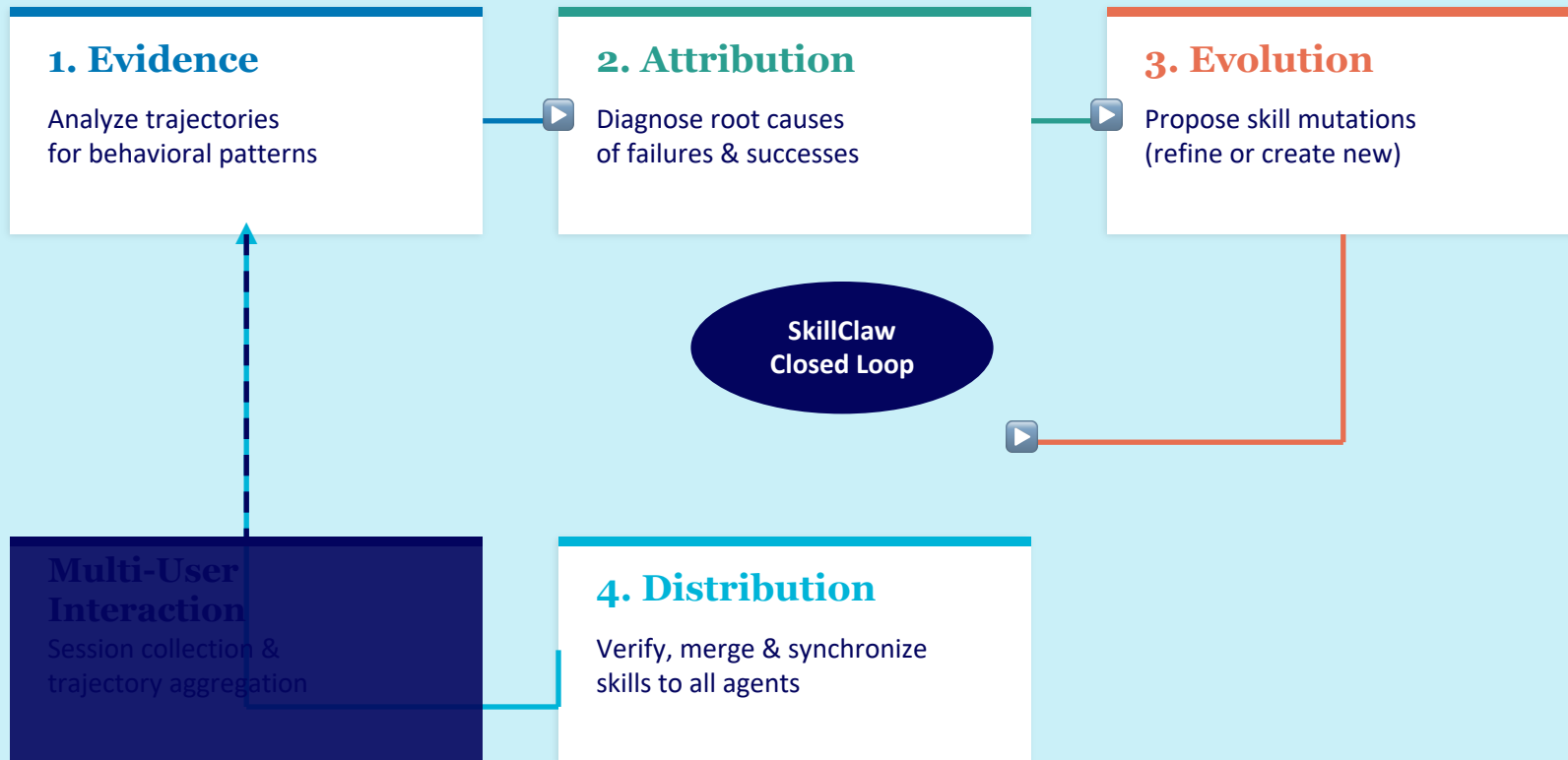
Evolving Skills (SkillClaw)

- Continuously updated from usage
- Cross-user evidence drives improvement
- Solutions persist and propagate
- Failures become evolution signals
- System-level knowledge accumulates

Memory-based methods stay instance-bound. Skill-based methods stay static. SkillClaw bridges both gaps.

Closed-Loop Pipeline

From Multi-User Interaction to Shared Skill Evolution



Agentic Evolver Pipeline

Evidence → Attribution → Evolution in Three Stages

1

Evidence

Analyze Trajectories

- Aggregate session trajectories by referenced skill
- Identify behavioral patterns across users
- Preserve full action-feedback causal chains

2

Attribution

Diagnose Root Causes

- Diagnose why failures and successes occur
- Separate generalizable patterns from noise
- Distinguish skill flaws from environment issues

3

Evolution

Propose Mutations

- Propose skill mutations based on evidence
- Refine existing skills or create new ones
- Open-ended reasoning, not predefined rules

The evolver is an autonomous agent performing open-ended reasoning over interaction evidence — not predefined update rules.

Cross-User Trajectory Aggregation

Exposes Skill Behavioral Boundaries

Complementary Signals

Different users exercising the same skill under diverse contexts produce complementary views — revealing both conditions where it works and where it breaks.

Single-User Limitation

A single user rarely generates enough signal to separate a generalizable improvement from an idiosyncratic fix. Cross-user aggregation is essential.

Compatible Claw-Style Systems

OpenClaw

CoPaw

IronClaw

PicoClaw

ZeroClaw

NanoClaw

NemoClaw

User Sessions
(Multi-Agent)



Grouped by
Referenced Skill



Shared Evidence Base
for Agentic Evolver

Case Study: Multimodal Creative Pipeline

6 Days of Evolution Attempts — 1 of 6 Accepted

Day	Candidate Skill	Skill Function	Validator	Next-Day Action
1	validate-tmp-workspace-inputs	Check /tmp_workspace inputs & env setup	Accept	Upgrade to new best pool
2	multimodal-input-validation-and-setup	Multimodal input validation & output env init	Reject	Keep current best pool
3	multimodal-creative-task-pipeline	Multimodal creative pipeline (PDF/video/image)	Reject	Keep current best pool
4	multimodal-creative-task-pipeline (impr.)	Added image classification, visual gen, validation	Reject	Keep current best pool
5	pipeline (impr.) + validate-required-input-files	Creative pipeline + per-file fail-fast validation	Reject	Keep current best pool
6	multimodal-creative-task-pipeline (cand.)	Extended PDF-to-poster / doc-to-visual paths	Reject	Not admitted to next cycle

Conservative verification prevents regressions. Only 1 of 6 candidates was accepted over 6 days, illustrating that the validator acts as a quality gate — rejecting incremental changes that do not demonstrably improve the skill pool.



= Accepted



= Rejected

Case Study: Git Push Auth-Fallback

Iterative Refinement Over 6 Days — 3 of 6 Accepted

Day	Candidate Skill	Change Summary	Validator	Next-Day Action
1	git-push-with-auth-fallback	Safe fallback instead of blocking on push failure	Accept	Add to Safety best pool
2	git-push-with-auth-fallback	Unified patch/bundle filenames, reduced inconsistency	Accept	Keep updated best pool
3	auth-fallback + git-clone-to-directory	Auth-alternative paths & secrets audit; fixed subshell pitfalls	Accept	Keep current best pool
4	(none)	Same-pool retest; validator confirmed current pool as best	Accept	Continue same best pool
5	git-push-with-auth-fallback	Added 'push hang treated as auth failure'; no improvement	Reject	Keep current best pool
6	git-push-with-auth-fallback	Added identity config & filename consistency; no improvement	Reject	Not admitted to next cycle

Iterative refinement works when evidence supports it. 3 of 6 days yielded accepted updates before reaching a plateau. Contrast with the creative pipeline (1/6): auth-fallback had clearer failure signals, enabling more successful mutations.



= Accepted



= Rejected

Key Takeaways

Three Properties That Distinguish SkillClaw

01 Collective Evolution

Knowledge from individual interactions contributes to a shared, continuously improving skill ecosystem. Users no longer rediscover solutions independently.

02 Fully Automatic

Skill evolution is driven by runtime interaction without manual curation or explicit user intervention. The system improves itself from usage.

03 Agentic Evolution Paradigm

Skill updates are produced through open-ended reasoning over evidence, not predefined update rules. The evolver adapts to new patterns autonomously.

Implications for Continuously Improving Agent Systems

SkillClaw demonstrates that multi-user agent ecosystems can self-improve when equipped with evidence-driven, agentic skill evolution — moving beyond static tooling toward living, adaptive skill repositories.